Shonal Gangopadhyay

Bloomington, MN | (952) 465-4190 | shonal.gangopadhyay@gmail.com | shonalgangopadhyay.com

Software engineer specializing in C++ and systems development, with experience designing robust, maintainable software across full SDLC workflows, including architecture, testing, and deployment. Proven ability to operate in structured engineering environments, deliver production-quality software, and lead complex technical projects. Exposure to machine learning systems and data pipelines in applied settings.

Experience

Software Engineering & Systems Teaching Specialist

Computer Science Department | University of Minnesota-Twin Cities | Minneapolis, MN

January 2022 – August 2025

- Designed and delivered production-oriented C++ software engineering courses serving 650+ students annually, emphasizing SOLID principles, design patterns, and maintainable architecture.
- Led multi-month team-based projects simulating real-world SDLC workflows (Waterfall → Agile/Scrum), including requirements engineering, UML design, implementation, testing, and iterative delivery.
- Taught and enforced industry-standard design practices, including abstraction, encapsulation, low coupling/high cohesion, and separation of concerns.
- Designed UML class diagrams using tools such as Lucidchart and draw.io to model large-scale object-oriented systems prior to implementation.
- Built CI-style pipelines using Python test harnesses and automated compilation workflows to validate thousands of C++ submissions.
- Reduced grading turnaround time by ~70% by implementing automated testing frameworks and validation harnesses.
- Integrated Git-based workflows, including branching, merging, code reviews, and static analysis, aligning coursework with modern collaborative development practices.
- Guided students in developing large-scale C++ systems involving memory management, polymorphism, dynamic allocation, and performance considerations.
- Served in leadership roles (Project Manager, Product Owner, Scrum Master), facilitating sprint planning, standups, and retrospectives while maintaining technical direction and delivery timelines.

Software Engineer Intern (Early-Stage Startup)

StudyHall AI | London, UK (Remote)

May 2022 – August 2022

- Developed backend services and data processing pipelines for adaptive educational systems, handling large, unstructured datasets.
- Collaborated with ML engineers to support model-driven features, including data preprocessing, feature extraction, and evaluation workflows.
- Contributed to systems leveraging machine learning models (PyTorch, scikit-learn) for content ranking and assessment logic.
- Designed and implemented features in a rapid prototyping environment, balancing correctness, scalability, and iteration speed.
- Participated in deployment and monitoring of services on AWS (EC2), focusing on reliability and performance.
- Operated in a high-autonomy startup environment, making architectural and implementation decisions with minimal predefined structure.

Education

M.S. Computer Science (AI/ML Focus), University of Minnesota – December 2023
B.A. Computer Science, University of Minnesota – May 2021

Technical Skills

Languages:
C++, Python, Java, TypeScript

Software Engineering & Systems:
Object-Oriented Design (OOD), SOLID Principles, Design Patterns (Factory, Singleton, Decorator, Facade, Observer, Strategy), UML (Lucidchart, draw.io), Memory Management (dynamic memory allocation, RAII, copy semantics), Polymorphism

Systems Programming:
Memory Management (dynamic allocation, copy semantics), C++ memory model, pointer semantics, stack vs heap behavior, Performance-aware programming (algorithmic complexity, memory/runtime tradeoffs)

Testing & Quality:
GoogleTest, Unit/Integration/Regression Testing, CI/CD (GitLab CI/CD, automated build/test pipelines), Static Analysis (compiler warnings, cpplint, code reviews)

Systems & Backend:
REST APIs, client-server architecture, HTTP protocols, Linux (Ubuntu), Bash scripting, Containerization (Docker), Networking fundamentals (ports, request/response lifecycle)

DevOps & Tools:
Git (GitHub, GitLab), GitLab CI/CD, Docker, AWS EC2, GCP, Jira

Machine Learning (Applied):
PyTorch, scikit-learn, Data Preprocessing, Model Evaluation, Feature Engineering

AI-Assisted Development:
AI-assisted development using LLMs for debugging, prototyping, and system design iteration

Projects

Shinobi Battlefield (Multiplayer Unity Game)

Unity (C#), Docker, WebGL, REST APIs, Client-Server Architecture
game.shonalgangopadhyay.com

- Designed and implemented a multiplayer game architecture with a dedicated server model, supporting 1–16 concurrent players over networked sessions.
- Built a client-server system using REST-style communication and containerized backend services, enabling remote connectivity via web deployment.
- Developed core gameplay systems including combat mechanics, state-driven enemy behavior, and round-based progression logic.
- Deployed the system using Docker and configured networking (port forwarding, DNS, reverse proxy) to support external player access.
- Iteratively debugged and refined full-stack interactions between Unity frontend and Python backend services.

Karate Judging System (Confidential Project)

Unity (C#), Python, Simulation Systems, Data Pipelines
judging.shonalgangopadhyay.com

- Designed and implemented a simulation-based evaluation system for real-time decision-making and event scoring in a structured rules-based environment.
- Developed data collection pipelines capturing high-frequency state and event data for downstream

analysis and model development.
- Built modular system architecture supporting extensible rule engines and future integration with machine learning models.
- Implemented visualization and debugging tools to analyze system behavior and validate decision outputs.
- Collaborated with stakeholders to align system behavior with domain-specific evaluation criteria.

Details available upon request under NDA

Karate Parent Portal Website
HTML, CSS, JavaScript, GitHub Pages, Responsive Design
karate.shonalgangopadhyay.com
- Designed and deployed a mobile-first informational website with structured navigation, responsive layouts, and consistent UI/UX patterns.
- Implemented reusable component-based layout (cards, sections, navigation systems) with externalized styling for maintainability.
- Configured domain routing and deployment via GitHub Pages and DNS forwarding (Squarespace).
- Built features such as sticky navigation, scroll tracking, and hierarchical content organization for improved usability.
- Iteratively refined layout, accessibility, and performance based on real user feedback.

Leadership
Head Instructor
National Karate | Multiple Locations, MN
January 2019 - Present
- Teach 400+ students, mentor black belt candidates, and develop high-performing competitors including state champions.

Director
Edina National Karate | Edina, MN
January 2014 – January 2019
- Managed operations of a 200+ student organization, including staffing, curriculum development, and business operations.
- Led hiring, training, and performance management while maintaining high instructional and operational standards.